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$$2(3z^2 - z + 1) + i(z + 1) = 0$$

$$\Rightarrow 6z^2 + (-2 + i)z + (2 + i) = 0$$

$$D = (-2 + i)^2 - 4 \cdot 6 \cdot (2 + i) = 3 - 4i - 24(2 + i) = 3 - 4i - 48 - 24i = -45 - 28i$$

$$\Rightarrow -45 - 28i = x^2 - y^2 + 2xyi$$

$$\left\{ \begin{array}{lcl} x^2 - y^2 & = & -45 \\ 2xy & = & -28 \end{array} \right. \Rightarrow \left\{ \begin{array}{lcl} x^2 - y^2 & = & -45 \\ x^2(-y)^2 & = & -196 \end{array} \right.$$

$$\Rightarrow \lambda^2 + 45\lambda - 196 = 0$$

$$D_\lambda = 45^2 - 4 \cdot 1 \cdot (-196) = 2025 + 784 = 2809 = 53^2$$

$$\lambda_1 = \frac{-45 + 53}{2} = 4 = x^2 \Rightarrow x = \pm 2$$

$$\lambda_2 = \frac{-45 - 53}{2} = -49 = -y^2 \Rightarrow y = \pm 7$$

$$\Rightarrow z_1 = \frac{2 - i + (2 - 7i)}{12} = \frac{4 - 8i}{12} = \frac{1}{3} - \frac{2}{3}i$$

$$\Rightarrow z_2 = \frac{2 - i - (2 - 7i)}{12} = \frac{6i}{12} = \frac{1}{2}i$$

References